A report on

Sliding Mode Baed Control for Trajectory Tracking of a Quadrotor under Single Rotor Fault

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BANGALORE

2026

Abstract

Quadrotor unmanned aerial vehicles (UAVs) are widely used in aerial robotics due to their agility and mechanical simplicity. However, their stability relies on the coordinated operation of four independent rotors. A fault in one rotor, such as loss of effectiveness (LOE) or complete failure, significantly degrades the controllability of the system and may lead to instability during flight. This report investigates a sliding mode control (SMC) strategy for trajectory tracking of a six degree-of-freedom (6-DOF) quadrotor in the presence of a single rotor fault. The objective is to design a robust control law that enables the quadrotor to follow a desired trajectory despite the loss of thrust effectiveness in one actuator. A control framework based on the nonlinear dynamics of the quadrotor will be developed using sliding mode control to ensure robust trajectory tracking. The proposed controller will be evaluated through simulations of the nonlinear quadrotor model under rotor fault conditions. The results will illustrate the capability of the SMC-based controller to maintain stable flight and achieve satisfactory trajectory tracking performance despite actuator degradation.

Contents

Chapter 1

Introduction

Unmanned aerial vehicles (UAVs), particularly quadrotor systems, have attracted significant attention in recent years due to their simple mechanical structure, vertical take-off and landing capability, and high maneuverability. These characteristics make quadrotors suitable for a wide range of applications such as surveillance, inspection, environmental monitoring, search and rescue, and logistics operations. Despite their advantages, quadrotors are inherently nonlinear, underactuated, and strongly coupled dynamical systems, which makes control design a challenging task. Achieving stable flight and accurate trajectory tracking requires the coordinated operation of four independent rotors that generate the thrust and torques necessary to control the vehicle motion.

Despite these advantages, quadrotors remain highly sensitive to actuator faults because their flight stability relies heavily on the coordinated operation of all four rotors. Rotor faults are particularly critical because they directly affect the thrust generation mechanism of the vehicle. A loss of effectiveness (LOE) in rotor or complete rotor failure significantly degrades the controllability of the system and may lead to instability during flight. In particular, the loss of a single rotor prevents independent control of roll, pitch, yaw, and thrust simultaneously, which may lead to instability and ultimately a crash if appropriate fault-tolerant control strategies are not employed. Research has shown that stable flight may still be achievable after rotor failure by sacrificing control over certain states, such as the yaw angle, and allowing the vehicle to spin around its vertical axis while maintaining control of the remaining critical states [1].

Several studies have investigated the behavior and controllability of quadrotors under rotor failure conditions. Mueller and D'Andrea has demonstrated that even after the complete loss of one or more propellers, a quadrotor can maintain controlled motion by spinning about a fixed axis and adjusting thrust to regulate position [2]. By tilting this rotational axis, translational motion can still be achieved, enabling the vehicle to maintain altitude and limited maneuverability despite actuator loss. These results highlight the possibility of achieving degraded but controllable flight under severe actuator faults.

In addition to actuator failures, quadrotor systems are often affected by modeling uncertainties, external disturbances, and parameter variations, all of which can degrade control performance. To address uncertainties and disturbances in nonlinear systems, sliding mode control (SMC) has been widely used due to its robustness properties and strong disturbance rejection capability. Sliding mode control forces the system trajectories to evolve on a predefined sliding surface, thereby ensuring robustness with respect to model uncertainties and external disturbances. In recent years, sliding mode based fault–tolerant control strategies have been explored for quadrotor systems. For example, Zhao *et al.* proposed a fault observer based adaptive sliding mode control strategy for quadrotors subject to actuator faults and disturbances, demonstrating improved tracking performance and robustness [3].

Motivated by existing research, this project investigates the design of a sliding mode control strategy for trajectory tracking of a quadrotor under a single rotor fault. The objective is to develop a control framework capable of maintaining stable flight and tracking desired trajectories despite the loss of one actuator. The proposed approach leverages the robustness properties of SMC to compensate for the underactuated dynamics introduced by rotor failure while ensuring accurate tracking performance.

The structure of this report is organized as follows. First, the dynamic model of the quadrotor system is presented, followed by the control allocation and way to induce the fault in the dynamics. Next, the problem formulation is described.

Chapter 2

System Description

2.1 System Description

The system considered in this project is a quadrotor unmanned aerial vehicle modeled as a rigid body with six degrees of freedom (6-DOF). The quadrotor consists of four rotors mounted at the ends of a cross-shaped frame, each producing an upward thrust force and an associated reaction torque. The vehicle motion is described using two coordinate frames: an inertial frame $\mathcal{I}$ fixed to the earth and a body-fixed frame $\mathcal{B}$ attached to the center of mass of the quadrotor.

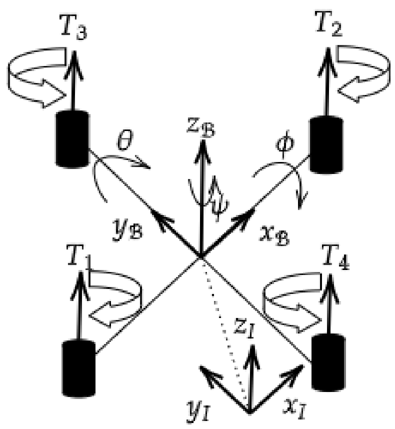

Figure 2.1: Configuration of Quadrotor

The state of the quadrotor is represented by the vector

$$x = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}]^T$$

where (x, y, z) denote the position of the quadrotor in the inertial frame, (ϕ, θ, ψ) represent the roll, pitch, and yaw angles respectively, and the remaining states represent the corresponding linear and angular velocities.

Integral error states are also included in the simulation model to support integral action in the controller.

2.1.1 Quadrotor Dynamic Model

The translational dynamics of the quadrotor are derived from Newton's second law and are expressed as

$$m\ddot{p} = Rf_T - mg$$

where m is the mass of the quadrotor, $p = [x, y, z]^T$ is the position vector, R is the rotation matrix from body frame to inertial frame, f_T is the total thrust generated by the four rotors, and g is gravitational acceleration.

Expanding the translational equations yields

$$\ddot{x} = \frac{T}{m}(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi) \tag{2.1}$$

$$\ddot{y} = \frac{T}{m}(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi) \tag{2.2}$$

$$\ddot{z} = \frac{T}{m}(\cos\phi\cos\theta) - g \tag{2.3}$$

where T represents the total thrust generated by the rotors.

The rotational dynamics are described using Euler's rigid body equations

$$I\dot{\omega} + \omega \times (I\omega) = \tau$$

where $I = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$ is the inertia matrix, $\omega = [\dot{\phi}, \dot{\theta}, \dot{\psi}]^T$ is the angular velocity vector, and $\tau = [\tau_\phi, \tau_\theta, \tau_\psi]^T$ represents the control torques.

The corresponding rotational acceleration equations are

$$\ddot{\phi} = \frac{I_{yy} - I_{zz}}{I_{xx}}\dot{\theta}\dot{\psi} + \frac{\tau_\phi}{I_{xx}} \tag{2.4}$$

$$\ddot{\theta} = \frac{I_{zz} - I_{xx}}{I_{yy}}\dot{\phi}\dot{\psi} + \frac{\tau_\theta}{I_{yy}} \tag{2.5}$$

$$\ddot{\psi} = \frac{I_{xx} - I_{yy}}{I_{zz}}\dot{\phi}\dot{\theta} + \frac{\tau_\psi}{I_{zz}} \tag{2.6}$$

2.1.2 Control Inputs and Allocation

The control inputs to the system are defined as

$$U = [u_1, u_2, u_3, u_4]^T$$

where

$$u_1 = T$$

is the total thrust, and

$$u_2 = \tau_\phi, \quad u_3 = \tau_\theta, \quad u_4 = \tau_\psi$$

represent the roll, pitch, and yaw torques respectively.

These inputs are generated by the individual rotor thrusts T_1, T_2, T_3, T_4. The relationship between rotor thrusts and control inputs is expressed through the control allocation matrix

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ \frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & -\frac{l}{\sqrt{2}} & \frac{l}{\sqrt{2}} \\ d & -d & d & -d \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \tag{2.7}$$

where l denotes the arm length and d is the torque-to-thrust coefficient.

2.1.3 Rotor Fault Model

To model actuator faults, the thrust generated by rotor 1 is scaled by an effectiveness parameter α. The fault model is given by

$$T_1^{fault} = \alpha T_1 \tag{2.8}$$

where $0 \leq \alpha \leq 1$. A value $\alpha = 1$ represents normal operation, while $\alpha = 0$ corresponds to complete rotor failure.

When rotor 1 fails completely, the system operates using the remaining three rotors and the control allocation matrix is reduced accordingly.

Using this dynamic model, the objective of the control system is to enable the quadrotor to track a desired trajectory while maintaining stability even in the presence of actuator faults.

Table 2.1: Notation used in the quadrotor dynamic model

Symbol	**Description**
x, y, z	Position of the quadrotor in the inertial frame (m)
ϕ, θ, ψ	Roll, pitch, and yaw angles (rad)
$\dot{x}, \dot{y}, \dot{z}$	Linear velocities (m/s)
$\dot{\phi}, \dot{\theta}, \dot{\psi}$	Angular velocities (rad/s)
m	Mass of the quadrotor (kg)
g	Gravitational acceleration (m/s^2)
I_{xx}, I_{yy}, I_{zz}	Moments of inertia (kg m^2)
p	Position vector $[x\ y\ z]^T$
R	Rotation matrix from body to inertial frame
ω	Angular velocity vector
T	Total thrust (N)
T_1, T_2, T_3, T_4	Individual rotor thrusts (N)
l	Arm length (m)
d	Torque-to-thrust coefficient
$\tau_\phi, \tau_\theta, \tau_\psi$	Roll, pitch, yaw torques (Nm)
u_1	Total thrust control input
u_2, u_3, u_4	Roll, pitch, yaw torque inputs
U	Control input vector
A	Control allocation matrix
α	Rotor effectiveness factor
T_1^{fault}	Effective thrust of rotor 1 under fault

Chapter 3

Problem Statement

The quadrotor is a nonlinear, underactuated aerial vehicle with six degrees of freedom (three translational and three rotational) and four control inputs generated by the rotors. The translational motion of the vehicle determines its position in the inertial frame, while the rotational motion determines its orientation described by roll (ϕ), pitch (θ), and yaw (ψ) angles.

The dynamic model of the quadrotor, derived using the Newton–Euler formulation in the previous section, can be expressed in the general nonlinear form

$$\dot{x} = f(x) + g(x)u$$

where $x \in \mathbb{R}^n$ represents the system state vector consisting of position, velocity, attitude angles, and angular velocities, $u \in \mathbb{R}^4$ denotes the control inputs corresponding to the collective thrust and control torques.

In practical flight scenarios, actuator faults may occur due to motor degradation, propeller damage, or electronic failures. A common type of actuator fault is the loss of effectiveness (LOE) in a rotor, where the generated thrust becomes smaller than the commanded value. In the extreme case, the rotor may completely fail and produce no thrust. Such faults directly affect the control authority of the quadrotor and can severely degrade its stability and trajectory tracking performance.

In this work, a single rotor fault is considered. The fault is modeled as a reduction in the effectiveness of the thrust generated by one rotor. If T_i denotes the nominal thrust produced by the i^{th} rotor, the faulty thrust can be expressed as

$$T_i^{\text{fault}} = \alpha_i T_i$$

where $0 \leq \alpha_i \leq 1$ represents the rotor effectiveness factor. A value $\alpha_i = 1$ corresponds to normal operation, $0 < \alpha_i < 1$ represents partial loss of effectiveness, and $\alpha_i = 0$ corresponds to complete rotor failure.

The presence of such actuator faults alters the input distribution of the quadrotor and introduces additional uncertainty into the control system. Consequently, the controller must maintain stability and tracking performance despite this degraded actuation. The primary control objectives are:

- Stabilization of the quadrotor with the presence of fault in rotor.
- Tracking of desired trajectories.
- Maintaining robustness against model uncertainties and disturbances.

To achieve these objectives, a sliding mode control strategy will be developed to provide robustness against model uncertainties, disturbances, and actuator degradation. The controller will be designed based on the nonlinear quadrotor dynamics presented earlier, and its effectiveness will be evaluated through numerical simulations of trajectory tracking under rotor fault conditions.

Chapter 4

Review of Control Methodologies

This chapter presents the control methodologies investigated for quadrotor trajectory tracking under actuator fault conditions.

4.1 PID Control

Proportional-Integral-Derivative (PID) control is one of the most widely used classical control strategies. It computes the control input as

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \dot{e}(t) \tag{4.1}$$

where $e(t)$ is the tracking error.

Although PID controllers are simple and effective for nominal conditions, they lack robustness against nonlinearities, disturbances, and actuator faults. In the presence of rotor faults, the PID controller fails to compensate for the loss of control authority.

4.2 Sliding Mode Control (SMC)

Sliding Mode Control is a nonlinear control technique known for its robustness. A sliding surface is defined as

$$s = \dot{e} + \lambda e \tag{4.2}$$

The control objective is to drive the system states to the sliding surface and maintain them on it. The control law typically consists of a continuous and a discontinuous component:

$$u = u_{eq} + u_{sw} \tag{4.3}$$

While SMC provides robustness against disturbances and model uncertainties, it assumes full control authority. Therefore, its performance degrades significantly in the presence of actuator faults.

Chapter 5

Fault Modeling and Analysis

This chapter presents the actuator fault modeling strategy used in this work and analyzes its impact on quadrotor control performance.

5.1 Actuator Fault Model

A loss-of-effectiveness (LOE) fault is considered in one rotor. The fault is modeled as a scaling of the thrust generated by the affected rotor:

$$T_1^{\text{actual}} = \alpha T_1, \quad 0 < \alpha \leq 1 \tag{5.1}$$

where α represents the actuator effectiveness. A value of $\alpha = 1$ corresponds to normal operation, while lower values indicate reduced thrust capability.

5.2 Control Allocation Framework

The control inputs of the quadrotor are given by

$$U = [u_1 \;\; u_2 \;\; u_3 \;\; u_4]^T \tag{5.2}$$

where u_1 is the total thrust and u_2, u_3, u_4 are roll, pitch, and yaw torques respectively.

These inputs are mapped to individual rotor thrusts using the control allocation matrix:

$$T = A^{\dagger} U \tag{5.3}$$

where $A^{\dagger}$ denotes the pseudo-inverse of the allocation matrix.

In the presence of actuator fault, the thrust vector is modified as:

$$T_{\text{actual}} = \begin{bmatrix} \alpha T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \tag{5.4}$$

The actual forces and torques applied to the system are then computed as:

$$U_{\text{actual}} = A\, T_{\text{actual}} \tag{5.5}$$

5.3 Effect of Fault on Control Inputs

Due to the presence of the fault, the actual control input differs from the commanded input:

$$U_{\text{actual}} \neq U \tag{5.6}$$

This mismatch introduces an implicit disturbance in the system. Expanding the above relation:

$$U_{\text{actual}} = U + \Delta U \tag{5.7}$$

where ΔU represents the error induced by the actuator fault.

This disturbance directly affects the system dynamics and reduces the effective control authority of the quadrotor.

5.4 Simulation Setup

The performance of the system is evaluated using an elliptical trajectory defined as:

$$x(t) = 4\cos(0.3t) \tag{5.8}$$

$$y(t) = 1\sin(0.3t) \tag{5.9}$$

$$z(t) = 1.5 \tag{5.10}$$

A cascaded PID controller is used for trajectory tracking. The fault is introduced in rotor 1 with different values of actuator effectiveness.

5.5 Results and Observations

5.5.1 Nominal Case ($\alpha = 1$)

Under normal operating conditions, the quadrotor accurately tracks the desired trajectory. The position states closely follow the reference signals, and the attitude states stabilize quickly. The control inputs and rotor thrusts remain bounded and smooth.

5.5.2 Moderate Fault ($\alpha = 0.5$)

When the actuator effectiveness is reduced, the system exhibits noticeable degradation in performance. The position tracking error increases, particularly in the horizontal plane. However, the controller is able to maintain overall stability and the system converges to the desired trajectory with reduced accuracy.

The mismatch between commanded and actual control inputs leads to transient oscillations in both attitude and thrust signals.

5.5.3 Severe Fault ($\alpha = 0.2$)

For a severe fault condition, the degradation becomes more pronounced. The quadrotor exhibits larger tracking errors and slower convergence. The reduction in thrust from the faulty rotor significantly limits the system's ability to generate the required control torques.

Despite these limitations, the system remains stable due to feedback control, but the tracking performance is substantially reduced.

5.6 Discussion

The results demonstrate that actuator faults introduce a mismatch between the commanded and realized control inputs, which acts as a disturbance in the system. As the level of fault increases, the available control authority decreases, leading to degraded tracking performance.

Since the controller does not explicitly account for actuator faults, it attempts to compensate for the error through feedback, resulting in increased control effort and oscillations. This highlights the need for fault-aware control strategies that incorporate actuator effectiveness into the control design.

Chapter 6

Simulation Studies

This chapter presents simulation results for trajectory tracking of a quadrotor under actuator fault conditions using two control strategies: cascaded PID control and Sliding Mode Control (SMC).

6.1 Simulation Setup

The quadrotor is required to track an elliptical trajectory defined as:

$$x(t) = 4\cos(0.3t) \tag{6.1}$$

$$y(t) = 1\sin(0.3t) \tag{6.2}$$

$$z(t) = 1.5 \tag{6.3}$$

The trajectory represents a horizontal ellipse with a semi-major axis of 4 m and semi-minor axis of 1 m. The altitude is maintained constant.

A single rotor fault is introduced in rotor 1 using an effectiveness parameter α. Three cases are considered:

- Nominal case: $\alpha = 1$
- Moderate fault: $\alpha = 0.5$
- Severe fault: $\alpha = 0.2$

6.2 PID Controller Performance

6.2.1 Nominal Case ($\alpha = 1$)

The PID controller achieves accurate trajectory tracking under nominal conditions. The position states closely follow the reference trajectory, and the tracking errors converge rapidly to zero. The attitude responses are smooth with minimal oscillations.

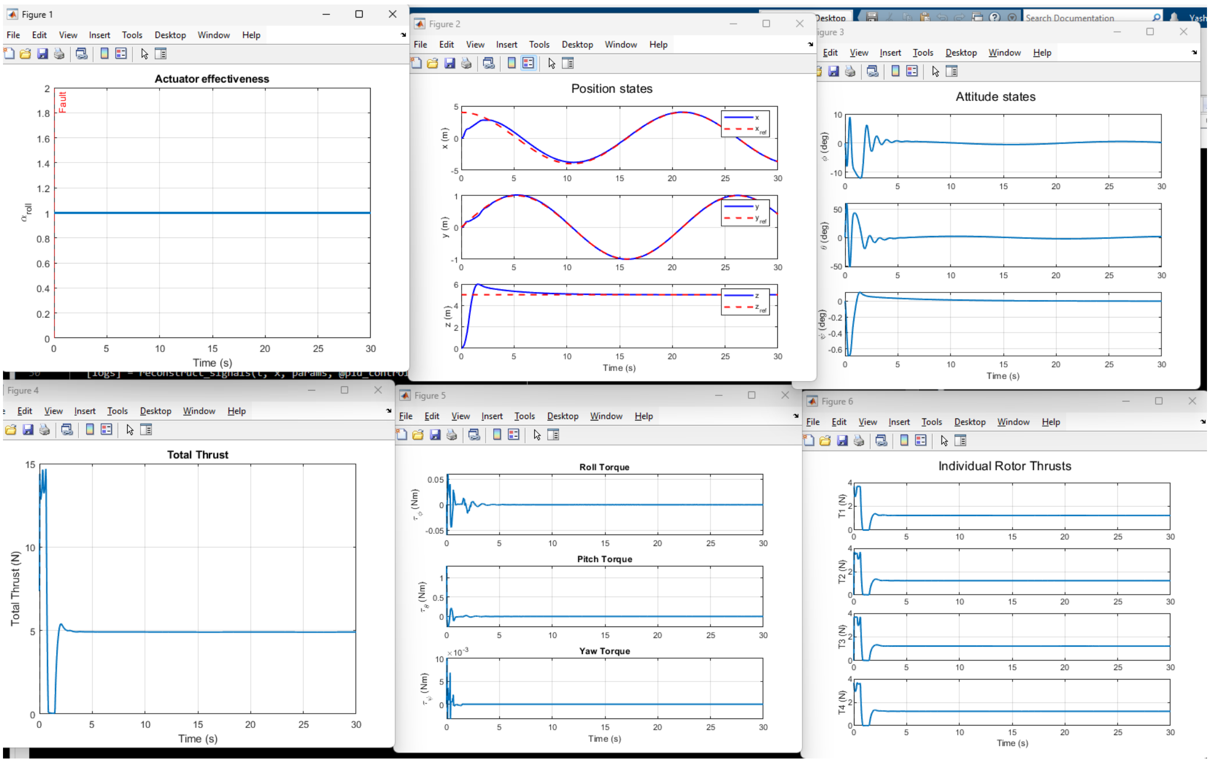

Figure 6.1: PID Case 1

6.2.2 Moderate Fault ($\alpha = 0.5$)

When a moderate fault is introduced, the PID controller exhibits noticeable degradation. The tracking error increases, particularly in the lateral motion (y-axis). Oscillations appear in the position error, indicating reduced stability margins.

The control inputs show increased effort as the controller attempts to compensate for the loss of thrust in the faulty rotor.

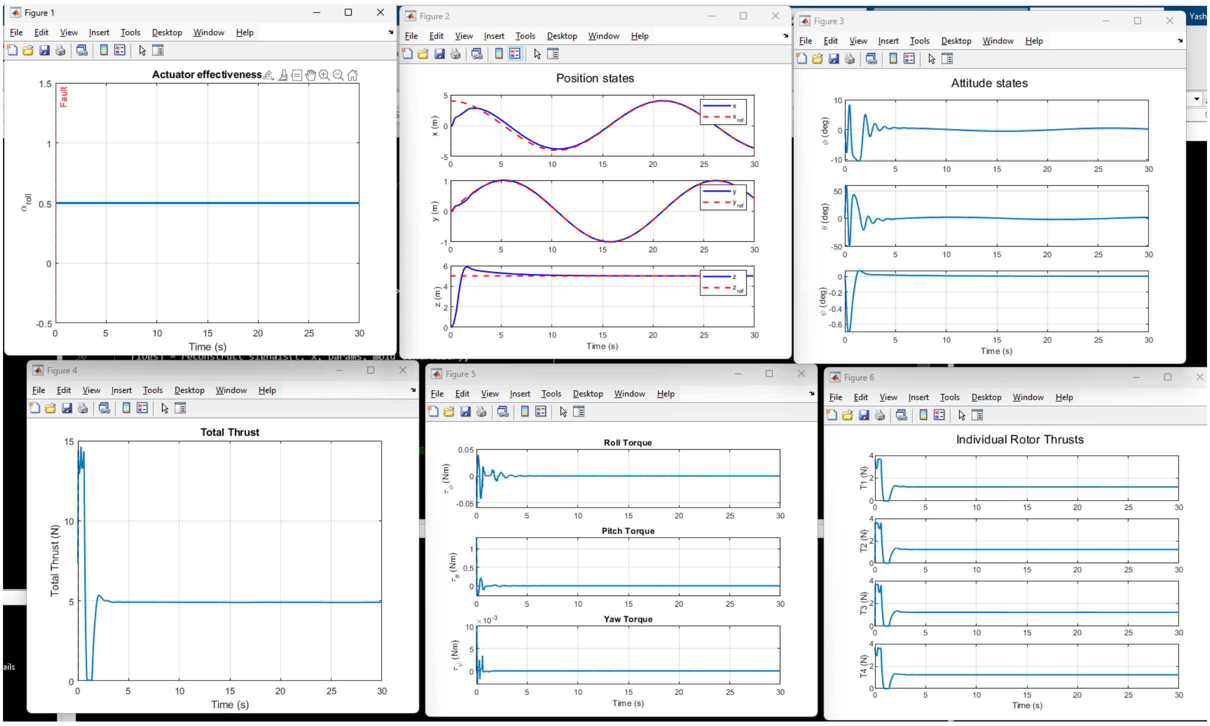

Figure 6.2: PID Case 2

6.2.3 Severe Fault ($\alpha = 0.2$)

Under severe fault conditions, the PID controller fails to maintain proper trajectory tracking. Significant steady-state errors appear, especially in altitude (z) and lateral motion. The system struggles to generate sufficient thrust and control torques, leading to degraded performance.

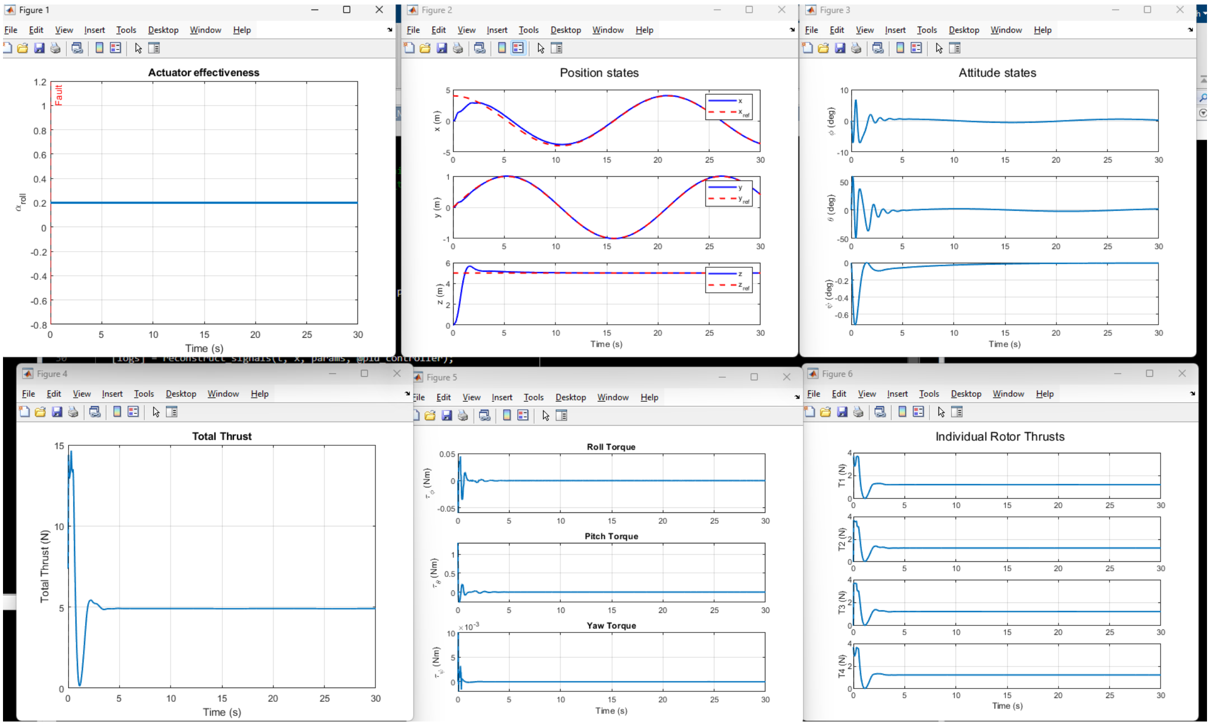

Figure 6.3: PID Case 3

6.3 Sliding Mode Control Performance

6.3.1 Nominal Case ($\alpha = 1$)

Under nominal conditions, the Sliding Mode Controller achieves accurate trajectory tracking with fast convergence. The tracking errors quickly reach zero, and the system exhibits strong robustness to initial transients.

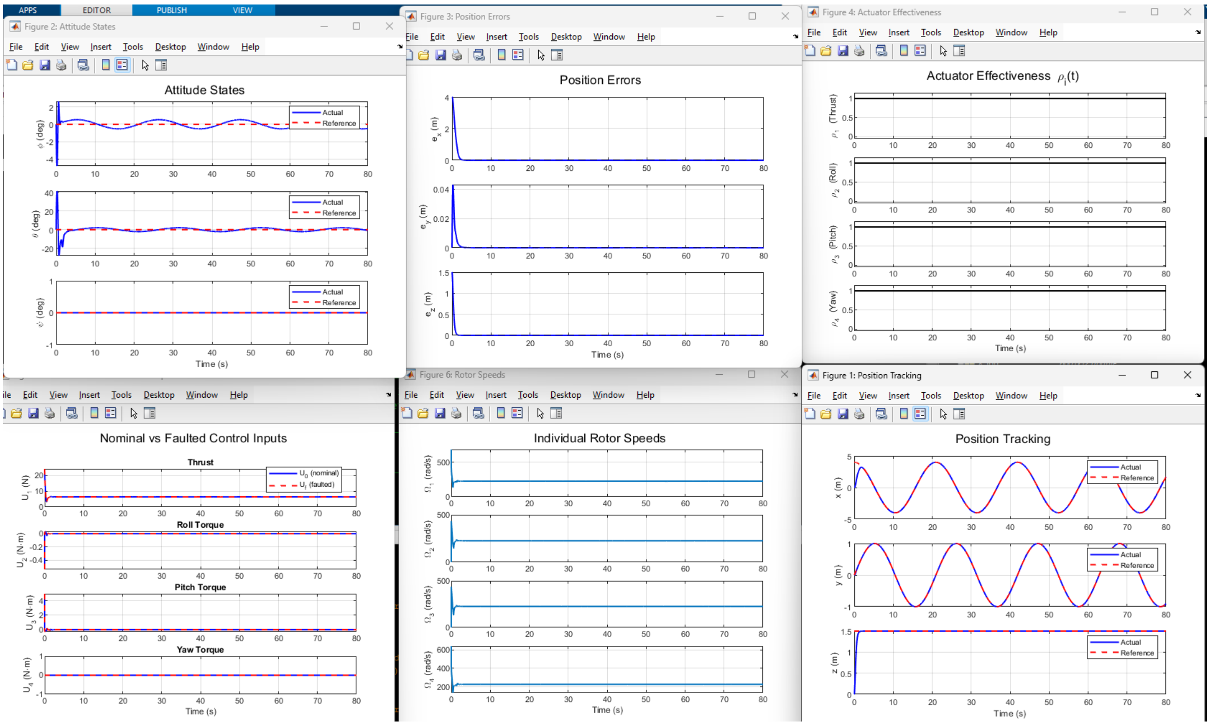

Figure 6.4: SMC Case 1

6.3.2 Moderate Fault ($\alpha = 0.5$)

With a moderate fault, SMC demonstrates improved robustness compared to PID. The trajectory tracking remains stable, and the tracking error is significantly lower than that observed with PID control.

Although small oscillations are present due to the discontinuous control action, the overall performance remains satisfactory.

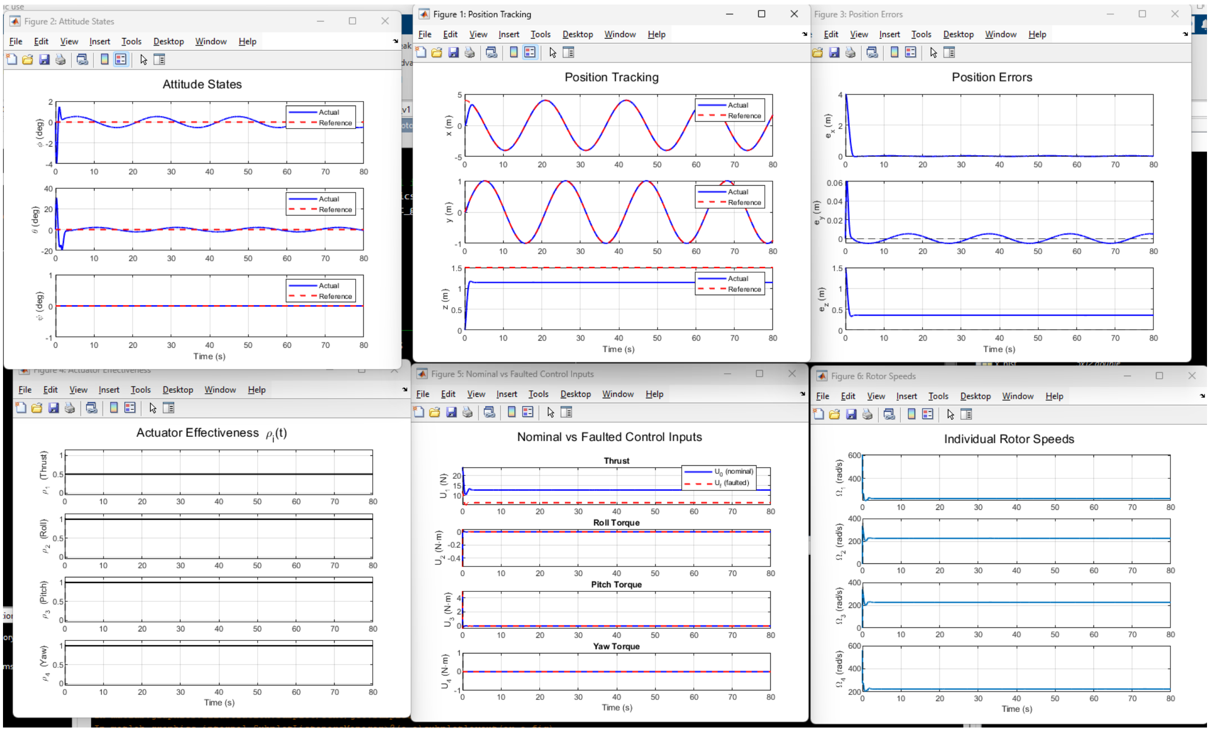

Figure 6.5: SMC Case 2

6.3.3 Severe Fault ($\alpha = 0.2$)

For severe faults, SMC maintains stability better than PID but exhibits noticeable degradation in tracking performance. The position errors increase, particularly in the horizontal plane, due to reduced control authority.

However, unlike PID, the system does not exhibit large steady-state divergence and remains bounded.

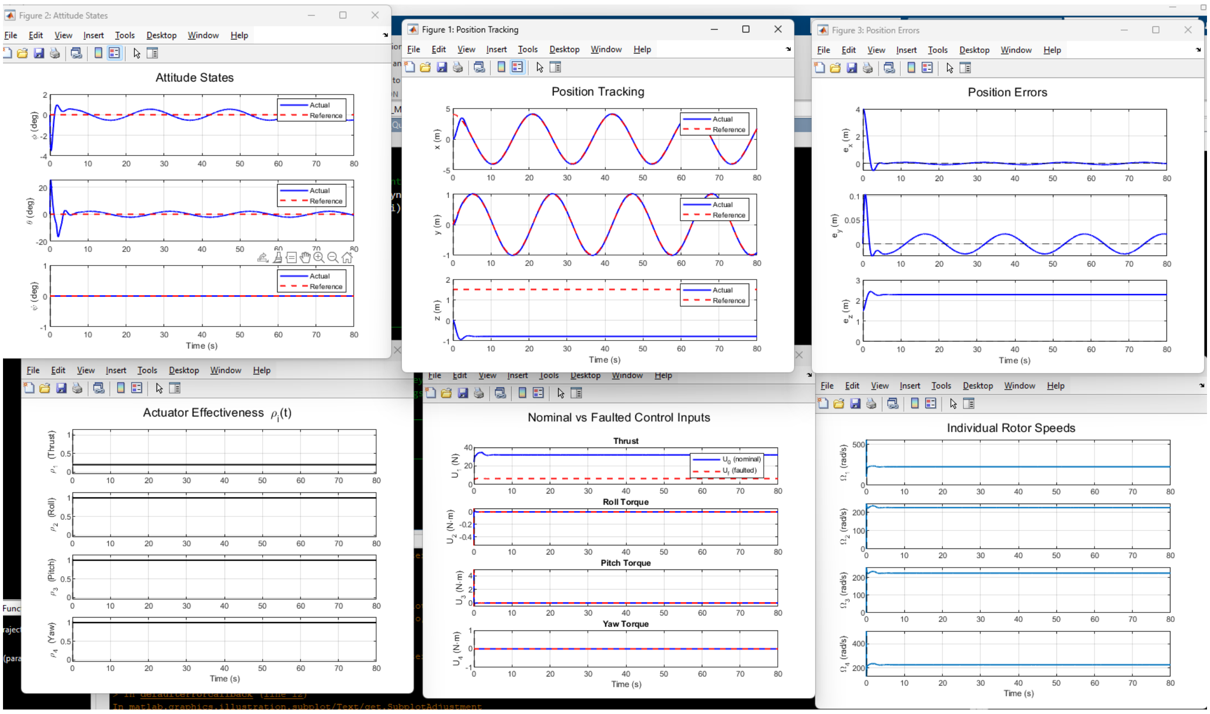

Figure 6.6: SMC Case 3

6.4 Comparative Analysis

The comparative results highlight the following observations:

- Both controllers perform well under nominal conditions.
- Under actuator fault, PID performance degrades significantly, especially for severe faults.
- SMC demonstrates improved robustness due to its ability to handle disturbances and uncertainties.
- The fault introduces a mismatch between commanded and actual inputs, which is better handled by SMC.

6.5 Key Observations

- Actuator faults reduce effective control authority, leading to increased tracking error.
- PID control lacks robustness against such faults and fails under severe degradation.
- SMC provides better disturbance rejection but does not fully compensate for actuator loss.

Chapter 7

Proposed Sliding Mode Control Framework

Based on the limitations observed in classical control approaches, a Sliding Mode Control (SMC) based framework is employed for robust trajectory tracking under actuator fault conditions.

7.1 Motivation

Simulation results with cascaded PID control demonstrate significant degradation in tracking performance under actuator faults. This is primarily due to the mismatch between commanded and actual control inputs introduced by the fault:

$$U_{\text{actual}} = U + \Delta U \tag{7.1}$$

Since PID control does not explicitly account for such disturbances, its performance deteriorates as the fault severity increases.

To address this issue, a robust nonlinear control strategy is required. Sliding Mode Control is well-suited for this purpose due to its inherent robustness to matched disturbances and model uncertainties.

7.2 Control Structure

A cascade SMC architecture is implemented, consisting of:

- Outer loop: position control (x, y, z)
- Inner loop: attitude control (ϕ, θ, ψ)

The controller generates virtual control inputs:

$$U = [U_1 \;\; U_2 \;\; U_3 \;\; U_4]^T \tag{7.2}$$

where U_1 is the total thrust and U_2, U_3, U_4 are control torques.

7.3 Sliding Surface Design

For each controlled state, a sliding surface is defined as:

$$s = \dot{e} + ce \tag{7.3}$$

where e is the tracking error and c is a positive constant.

For example, the altitude sliding surface is:

$$s_z = \dot{e}_z + c_z e_z \tag{7.4}$$

Similar surfaces are defined for position and attitude states.

7.4 Control Law

The SMC control law consists of three components:

$$U = U_{\text{eq}} + U_{\text{sw}} + U_{\text{lin}} \tag{7.5}$$

7.4.1 Equivalent Control

The equivalent control term cancels known system dynamics and ensures motion along the sliding surface:

$$U_{\text{eq}} \sim \text{model-based compensation} \tag{7.6}$$

For example, the altitude control input is designed as:

$$U_1 = \frac{m}{\cos\phi\cos\theta}\left(g + \text{dynamic terms}\right) \tag{7.7}$$

7.4.2 Switching Control

The switching term enforces convergence to the sliding surface:

$$U_{\text{sw}} = k_s\,\text{sat}(s) \tag{7.8}$$

A saturation function is used instead of the sign function to reduce chattering:

$$\text{sat}(s) = \begin{cases} \frac{s}{\delta}, & |s| < \delta \\ \text{sign}(s), & \text{otherwise} \end{cases} \tag{7.9}$$

7.4.3 Linear Term

A linear term is added to improve convergence:

$$U_{\text{lin}} = k_l s \tag{7.10}$$

7.5 Handling of Actuator Fault

The actuator fault introduces a mismatch between commanded and realized inputs:

$$U_{\text{actual}} \neq U \tag{7.11}$$

This mismatch acts as a disturbance in the system dynamics. Since Sliding Mode Control is inherently robust to matched disturbances, it compensates for this mismatch through the switching control term.

As a result, the system maintains stability and acceptable tracking performance even under actuator degradation.

7.6 Advantages of the Proposed Approach

- Robust to actuator faults and model uncertainties
- Improved tracking performance compared to PID control
- Reduced sensitivity to disturbances
- Modular cascade structure suitable for implementation

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